FISCAL POLICY TOOLS

Learning Outcome

- describe tools of fiscal policy, including their advantages and disadvantages

We now look at the nature of the fiscal tools available to a government. Government spending can take a variety of forms:

- **Transfer payments** are welfare payments made through the social security system and, depending on the country, include payments for state pensions, housing benefits, tax credits and income support for poorer families, child benefits, unemployment benefits, and job search allowances. Transfer payments exist to provide a basic minimum level of income for low-income households, and they also provide a means by which a government can change the overall income distribution in a society. Note that these payments are not included in the definition of GDP because they do not reflect a reward to a factor of production for economic activity. Also, they are not considered to be part of general government spending on goods and services.
- **Current government spending** involves spending on goods and services that are provided on a regular, recurring basis—including health, education, and defense. Clearly, such spending will have a big impact on a country's skill level and overall labor productivity.
- **Capital expenditure** includes infrastructure spending on roads, hospitals, prisons, and schools. This investment spending will add to a nation's capital stock and affect productive potential for an economy.

Government spending can be justified on both economic and social grounds:

- To provide such services as defense that benefit all citizens equally.
- For infrastructure capital spending (e.g., roads) to help a country's economic growth.
- To guarantee a minimum level of income for poorer people and hence redistribute income and wealth (e.g., welfare and related benefits).
- To influence a government's economic objectives of low inflation and high employment and growth (e.g., management of aggregate demand).
- To subsidize the development of innovative and high-risk new products or markets (e.g., alternative energy sources).

Government revenues can take several forms:

- **Direct taxes** are levied on income, wealth, and corporate profits and include capital gains taxes, national insurance (or labor) taxes, and corporate taxes. They also may include a local income or property tax for both individuals and businesses. Inheritance tax on a deceased's estate will have both revenue-raising and wealth-redistribution aspects.
- **Indirect taxes** are taxes on spending on a variety of goods and services in an economy—such as the excise duties on fuel, alcohol, and tobacco as well as sales (or value-added tax)—and often exclude health and education products on social grounds. In addition, taxes on gambling may be considered to have a social aspect in deterring such activity, while fuel duties will serve an environmental purpose by making fuel consumption and hence travel more expensive.

Taxes can be justified both in terms of raising revenues to finance expenditures and in terms of income and wealth redistribution policies. Economists typically consider four desirable attributes of a tax policy:

- Simplicity: This refers to ease of compliance by the taxpayer and enforcement by the revenue authorities. The final liability should be certain and not easily manipulated.
- Efficiency: Taxation should interfere as little as possible in the choices individuals make in the marketplace. Taxes affect behavior and should, in general, discourage work and investment as little as possible. A major philosophical issue among economists is whether tax policy should deliberately deviate from efficiency to promote "good" economic activities, such as savings, and discourage harmful ones, such as tobacco consumption. Although most would accept a limited role in guiding consumer choices, some will question if policy makers are equipped to decide on such objectives and whether there will be unwanted ancillary effects, such as giving tax breaks for saving among people who already save and whose behavior does not change.
- Fairness: This refers to the fact that people in similar situations should pay the same taxes ("horizontal equity") and that richer people should pay more taxes ("vertical equity"). Of course, the concept of fairness is really subjective. Still, most would agree that income tax rates should be progressive—that is, that households and corporations should pay proportionately more as their incomes rise. However, some people advocate "flat" tax rates, whereby all should pay the same proportion of taxable income.
- Revenue sufficiency: Although revenue sufficiency may seem obvious as a criterion for tax policy, there may be a conflict with fairness and efficiency. For example, one may believe that increasing income tax

Some Issues with Tax Policy

1. **Incentives.** Some economists believe that income taxes reduce the incentive to work, save, and invest and that the overall tax burden has become excessive. These ideas are often associated with supply-side economics and the US economist Arthur Laffer. A variety of income tax cuts and simplifications have taken place in the United States since 1981, and despite substantial controversy, some claim that work effort did rise (although tax cuts had little impact on savings). Similarly, some found that business investment did rise, while others claimed it was independent of such cuts.
2. **Fairness.** How do we judge the fairness of the tax system? One way is to calibrate the tax burden falling on different groups of people ranked by their income and to assess how changes in taxes affect these groups. Of course, this imposes huge data demands on investigators and must be considered incomplete. In the United States, the federal system is indeed highly progressive. Many countries use such methods to analyze the impact of tax changes on different income groups when they announce their annual fiscal policy plans.
3. **Tax reform.** There is continuous debate on reforming tax policy. Should there be a flat-rate tax on labor income? Should all investment be immediately deducted for corporate taxes? Should more revenue be sourced from consumption taxes? Should taxes be indexed to inflation? Should dividends be taxed when profits have already been subject to tax? Should estates be taxed at all? Many of these issues are raised in the context of their impact on economic growth.

Question Set

1. Which of the following is *not* a tool of fiscal policy?

A. A rise in social transfer payments
B. The purchase of new equipment for the armed forces
C. An increase in deposit requirements for the buying of houses

Hide Solution

Solution:

C is correct. Rises in deposit requirements for house purchases are intended to reduce the demand for credit for house purchases and hence would be considered a tool of monetary policy. This is a policy used actively in several countries and is under consideration by regulators in other countries to constrain house price inflation.

2. Which of the following is *not* an indirect tax?

A. Excise duty
B. Value-added tax
C. Employment taxes

Hide Solution

Solution:

C is correct. Both excise duty and value-added tax (VAT) are applied to prices, whereas taxes on employment apply to labor income and hence are not indirect taxes.

3. Which of the following statements is *most* accurate?

A. Direct taxes are useful for discouraging alcohol consumption.
B. Because indirect taxes cannot be changed quickly, they are of no use in fiscal policy.
C. Government capital spending decisions are slow to plan, implement, and execute and hence are of little use for short-term economic stabilization.

Hide Solution

Solution:

C is correct. Capital spending is much slower to implement than changes in indirect taxes; and indirect taxes affect alcohol consumption more directly than direct taxes.

The Advantages and Disadvantages of Different Fiscal Policy Tools

The different tools used to expedite fiscal policy as a means to try to put or keep an economy on a path of positive, stable growth with low inflation have both advantages and disadvantages:

Advantages

- Indirect taxes can be adjusted almost immediately after they are announced and can influence spending behavior instantly and generate revenue for the government at little or no cost to the government.

Disadvantages

- Direct taxes are more difficult to change without considerable notice, often many months, because payroll computer systems will have to be adjusted (although the announcement may well have a powerful effect on spending behavior more immediately). The same may be said for welfare and other social transfers.
- Capital spending plans take longer to formulate and implement, typically over a period of years. For example, building a road or hospital requires detailed planning, legal permissions, and implementation. This is often a valid criticism of an active fiscal policy and was widely heard during the US fiscal stimulus in 2009–2010. Such policies, however, do add to the productive potential of an economy, unlike a change in personal or indirect taxes. Of course, the slower the impact of a fiscal change, the more likely other exogenous changes will already be influencing the economy before the fiscal change kicks in.

These tools may have expectational effects at least as powerful as the direct effects. The announcement of future income tax rises a year ahead potentially could lead to reduced consumption immediately. Such delayed tax rises were a feature of the UK fiscal policy of 2009–2010; however, the evidence is anecdotal because spending behavior changed little until the delayed tax changes actually came into force.

We may also consider the relative potency of the different fiscal tools. Direct government spending has a far bigger impact on aggregate spending and output than income tax cuts or transfer increases; however, if the latter are directed at the poorest in society (basically, those who spend all their income), then this will give a relatively strong boost. Further discussion and examples of these comparisons are given in the section on the interaction between monetary and fiscal policy.

Modeling the Impact of Taxes and Government Spending: The Fiscal Multiplier

The conventional macroeconomic model has government spending, G, adding directly to aggregate demand, AD, and reducing it via taxes, T; these include both indirect taxes on expenditures and direct taxes on factor incomes. Further government spending is increased through the payment of transfer benefits, B, such as social security payments. Hence, the net impact of the government sector on aggregate demand is as follows:

$G - T + B$ = Budget surplus OR deficit.

Net taxes (NT; taxes less transfers) reduce disposable income (YD) available to individuals relative to national income or output (Y) as follows:

$YD = Y - NT = (1 - t)\ Y$,

where t is the **net tax rate**. Net taxes are often assumed to be proportional to national income, Y, and hence total tax revenue from net taxes is tY. If t = 20% or 0.2, then for every USD1 rise in national income, net tax revenue will rise by USD0.20 and household disposable income will rise by USD0.80.

The **fiscal multiplier** is important in macroeconomics because it tells us how much output changes as exogenous changes occur in government spending or taxation. The recipients of the increase in government spending will typically save a proportion $(1 - c)$ of each additional dollar of disposable income, where c is the **marginal propensity to consume** (MPC) this additional income. Ignoring income taxes, we can see that $\$c$ will, in turn, be spent by these recipients on more goods and services. The recipients of this $\$c$ also will spend a proportion c of this additional income (i.e., $\$c \times c$, or c-squared). This process continues with income and spending growing at a constant rate of c as it passes from hand to hand through the economy. This is the familiar geometric progression with constant factor c, where $0 < c < 1$. The sum of this geometric series is $1/(1 - c)$.

We define s as the **marginal propensity to save** (MPS), the amount saved out of an additional dollar of disposable income. Because $c + s = 1$, $s = 1 - c$.

Exhibit 9: Disposable Income, Saving, and the MPC

Income	Income Tax	Disposable Income	Consumption	Saving
USD100	USD20	USD80	USD72	USD8

In Exhibit 9, the MPC out of disposable income is 90 percent or 0.9 (72/80). The MPS is therefore 1 – 0.9 or 0.1.

For every dollar of new (additional) spending, total incomes and spending rises by USD$1/(1 - c)$. And because $0 < c < 1$, this must be > 1; this is the multiplier. If $c = 0.9$ (or individuals spend 90 percent of additions to income), then the multiplier = $1/(1 - 0.9) = 10$.

A formal definition of the multiplier would be the ratio of the change in equilibrium output to the change in autonomous spending that caused the change. This is a monetary measure, but because prices are assumed to be constant in this analysis, real and monetary amounts are identical. Given that fiscal policy is about changes in government spending, G, net taxes, NT, and tax rates, t, we can see that the multiplier is an important tool for calibrating the possible impact of policy changes on output. How can we introduce tax changes into the multiplier concept? We do this by introducing the idea of disposable income, YD, defined as income less income taxes net of transfers, $Y - NT$.

spending, say on road building, by an amount, G, then disposable income rises by $(1 - t)G$ and consumer spending by $c(1 - t)G$. Provided there are unused sources of capital and labor in the economy, this leads to a rise in aggregate demand and output; the recipients of this extra consumption spending will have $(1 - t)c(1 - t)G$ extra disposable income available and will spend c of it. This cumulative extra spending and income will continue to spread through the economy at a decreasing rate as $0 < c(1 - t) < 1$. The overall final impact on aggregate demand and output will effectively be the sum of this decreasing geometric series with the common ratio $c(1 - t)$, which sums to $1/[1 - c(1 - t)]$. This is known as the fiscal multiplier and is relevant to studies of fiscal policy as changes in G or tax rates will affect output in an economy through the value of the multiplier.

For example, if the tax rate is 20 percent, or 0.2, and the marginal propensity to consume is 90 percent, or 0.9, then the fiscal multiplier will be as follows: $1/[1 - 0.9(1 - 0.2)]$ or $1/0.28 = 3.57$. In other words, if the government raises G by USD1 billion, total incomes and spending rise by USD3.57 billion.

Discretionary fiscal policy will involve changes in these variables with a view to influencing Y.

The Balanced Budget Multiplier

If a government increases G by the same amount as it raises taxes, the aggregate output actually rises. Why is this?

Because the marginal propensity to consume out of disposable income is less than 1, for every dollar less in YD, spending falls only $\$c$. Hence, aggregate spending falls less than the tax rise by a factor of c. A balanced budget leads to a rise in output, which in turn leads to further rises in output and incomes through the multiplier effect.

Suppose an economy has an equilibrium output or income level of USD1,000 consisting of USD900 of consumption and USD100 of investment spending, which is fixed and not related to income. If government spending is set at USD200, financed by a tax rate of 20 percent (giving tax revenue of USD200), what will happen to output? First, additional government spending of USD200 will raise output by that amount. But taxes of USD200 will not reduce output by a similar amount if the MPC is less than 1. Suppose it is 0.9, and hence spending will fall only by 90 percent of USD200, or USD180. The initial impact of the balanced fiscal package on aggregate demand will be to raise it by USD200 – USD180 = USD20. This additional output, in turn, will lead to further increases in income and output through the multiplier effect.

Even though this policy involved a combination of government spending and tax increases that initially left the government's budget deficit/surplus unchanged, the induced rise in output will lead to further tax revenue increases and a further change in the budget position. Could the government adjust the initial change in spending to offset exactly the eventual total change in tax revenues? The answer is "yes," and we can ask what the effect will be on output of this genuinely balanced budget change. This balanced budget multiplier always takes the value unity.

EXAMPLE 1

Government Debt, Deficits, and Ricardo Equivalence

The total stock of government debt is the outstanding stock of IOUs issued by a government and not yet repaid. They are issued when the government has insufficient tax revenues to meet expenditures and has to borrow from the public. The size of the outstanding debt equals the cumulative quantity of net borrowing it has done, and the fiscal or budget deficit is added in the current period to the outstanding stock of debt. If the outstanding stock of debt falls, we have a negative deficit or a surplus.

If a government reduces taxation by USD10 billion one year and replaces that revenue with borrowing of USD10 billion from the public, will it have any real impact on the economy? The important issue is how people perceive that action: Do they recognize what will happen over time as interest and bond principal have to be repaid out of future taxes? If so, they may think of the bond finance as equivalent to delayed taxation finance; thus, the reduction in current taxation will have no impact on spending because individuals save more in anticipation of higher future taxes to repay the bond. This is called **Ricardian equivalence** after the economist David Ricardo. If people do not correctly anticipate all the future taxes required to repay the additional government debt, then they feel wealthier when the debt is issued and may increase their spending, adding to aggregate demand.

Whether Ricardian equivalence holds in practice is ultimately an empirical issue and is difficult to calibrate conclusively given the number of factors that are changing at any time in a modern economy.

Question Set

1. Which of the following is the *most likely* example of a tool of fiscal policy?

A. Public financing of a power plant
B. Regulation of the payment system
C. Central bank's purchase of government bonds

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A is correct. Public financing of a power plant could be described as a fiscal policy tool to stimulate investment.

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